SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 25

Code of Conduct

I Abdul Riyaz Ahammad (192525406) (Name / Reg. No.) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Abdul Riyaz Ahammad

Assessment Tasks
Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

Total Marks:

Defination: Cloud computing is the delivery of computing resources such as servers, storage, databases and software over the internet on demand

Four charcterstics

1. On-demand self-service: Users can access resource whenever needed without human assistance

2. Broad network access: Services are accesible through the internet using laptops, mobiles, and other devices

3. Resource Pooling: Providers share computing resources among multiple users

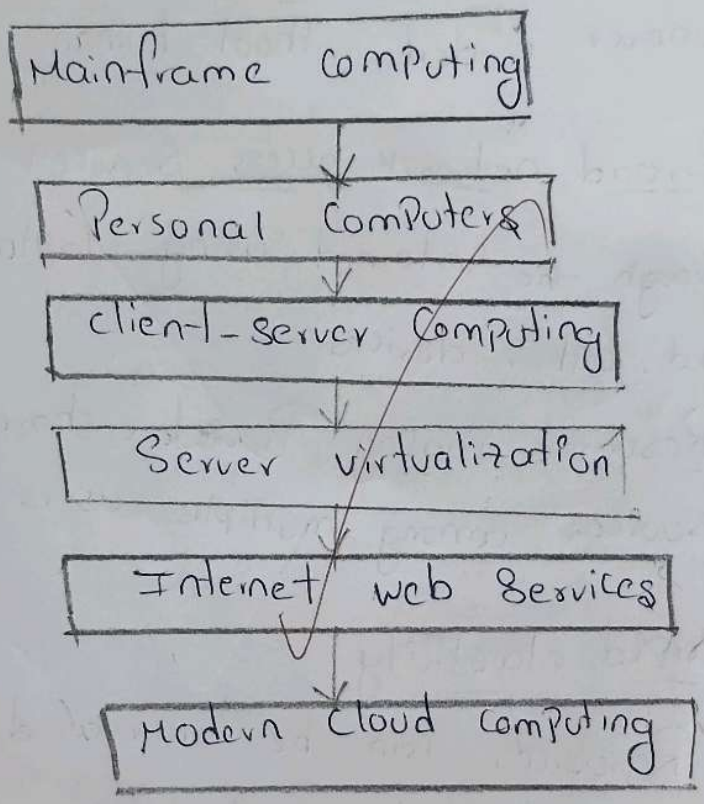
4. Rapid elasticity
   Resources can be increased or decreased quickly based on demand

Difference

Traditional on-Premise Computing requires organization to Purchase and maintain their own hardware, whereas cloud computing reduces upfront costs and Provides flexible resources

2) Mainframes enabled centralized computing, Personal computers brought individual access, and the internet enabled remote acess to cloud services

Flowchart



Advantages in hardware, virtualization, high speed Internet, and web technologies enabled modern cloud platforms such as Aws, Microsoft Azure, and Google cloud.

3) Role of Server Virtualization

Server virtualization divides one Physical Server into VMs. Each VM can run its own operating system and applications, improving resource utilization and reducing costs

Virtualization	cloud Computing
1. Technology that creates virtual resources	1. Service model that deliveries computing resources on internet
2. Runs multiple VMs on one Physical Server	2. Provides on-demand resources and software
3. Enables efficient hardware utilization	3. Provides scalability, self service and Pay-as-you-use services
4. Example:- VMware	4. Examples :- 1. Amazon web Services (AWS)

4)

* data centre is a facility containing servers, storage, systems, Networking equipment and supporting infrastructure

Role of Data centres

1. Data storage
stores applications, files and databases

2. Processing
Provides computing power to run cloud applications

3. Networking
Connects users to cloud resources through internet

4. Availability
Uses backup system and redudancy to reduce service

5. Security:
Protects data through access controls and Security mechanisms

6. Scalability: supports individual computing resources as demand increases

Model	Meaning	Benifits
Iaas	Infrastructure as a service	Provides virtual servers, Storage & Networks
Paas	Platform as a service	Provides a platform to develop and deploy
Saas	software as a Service	Provides ready-to-use software on internet
Xaas	everything as service	Deliveries various IT resources and service on demand

Key differences

- Iaas : User manages the operating system applications
- Paas : user focuses mainly on application development
- Saas : Users simply use Software Provider manages infrastructure & applications
- Xaas :- A broad team covering many services as needed

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